

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 24

Route Security Hub Findings to Email with EventBridge and SNS

Amazon EventBridge | Amazon SNS | AWS Security Hub | SecurityAudit

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Executive Summary

Created an **Amazon SNS** topic named **SecurityHubAlerts** in the **SecurityAudit** account, created an email subscription for alert delivery, and created an **Amazon EventBridge** rule named **SecurityHubFindings** to match imported Security Hub findings and send matched events to the SNS topic. This project turns the centralized Security Hub feed into a basic notification path for lab alerts.

PRODUCTION
CAVEAT

This is a lab notification pattern. Email is acceptable for low-volume learning environments, but production environments should route Security Hub findings to a SIEM, ticketing system, incident-response workflow, or severity-based automation.

Business Problem

Project 23 centralized Security Hub findings, but centralized visibility alone does not notify the operator when a new finding arrives. EventBridge can match Security Hub imported-finding events in near real time and route them to downstream targets. SNS provides a simple delivery path to email for this lab environment.

Objectives

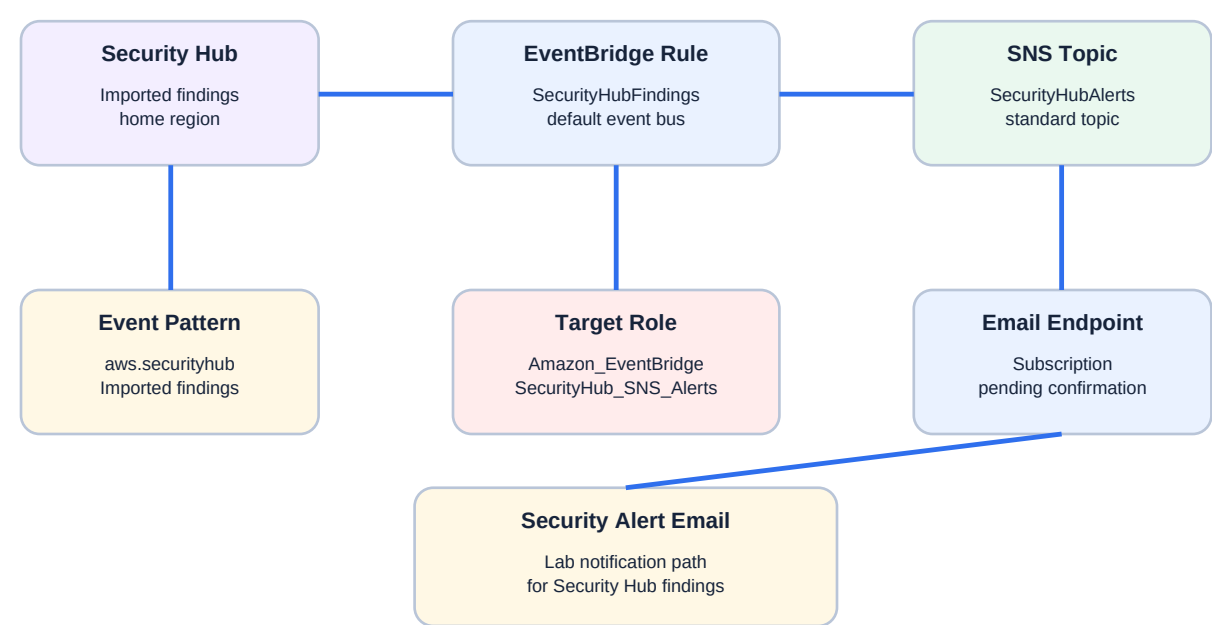
- Work from the SecurityAudit account instead of the management account.
- Create the SecurityHubAlerts SNS topic.
- Create an email subscription for the SecurityHubAlerts topic.
- Create an EventBridge rule named SecurityHubFindings.
- Use an event pattern that matches Security Hub imported findings.
- Send matched events to the SecurityHubAlerts SNS topic.
- Document that the email subscription was pending confirmation at evidence capture time.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Services	Amazon EventBridge, Amazon SNS, AWS Security Hub
Account Context	SecurityAudit
Region	us-west-2 / Oregon
SNS Topic	SecurityHubAlerts
Subscription Protocol	EMAIL
Subscription Status	Pending confirmation at capture time

EventBridge Rule	SecurityHubFindings
Event Bus	default
Rule Status	Enabled
Rule Type	Standard rule
Target	SecurityHubAlerts SNS topic
Event Pattern	source aws.securityhub and detail-type Security Hub Findings - Imported

Architecture



EventBridge captures Security Hub imported-finding events and forwards matched events to SNS email.

Figure 1 - Security Hub findings routed through EventBridge to SNS email notification.

Implementation Summary

1. Signed into IAM Identity Center and selected SecurityAudit with the SecurityFullAdmin access path.
2. Verified the region was Oregon / us-west-2, matching the Security Hub home region.
3. Opened Amazon SNS and created the SecurityHubAlerts topic.
4. Created an email subscription for the SecurityHubAlerts SNS topic.
5. Opened Amazon EventBridge and created a standard rule named SecurityHubFindings.
6. Configured the rule on the default event bus.

- 7. Built a custom event pattern for Security Hub imported findings.
- 8. Selected the SecurityHubAlerts SNS topic as the rule target.
- 9. Allowed EventBridge to use a role to publish matched events to SNS.
- 10. Reviewed and created the rule.

EVENT
PATTERN

The rule matches imported Security Hub findings using **source = aws.securityhub** and **detail-type = Security Hub Findings - Imported**.

Event Pattern

```
{
  "source": ["aws.securityhub"],
  "detail-type": ["Security Hub Findings - Imported"]
}
```

Routing Map

Finding source	Security Hub imported findings	Captures findings aggregated into the Security Hub home region.
Event match	EventBridge rule SecurityHubFindings	Filters the event stream to the desired Security Hub event type.
Notification target	SNS topic SecurityHubAlerts	Receives matched events from EventBridge.
Delivery endpoint	Email subscription	Sends lab alerts to the configured email endpoint after confirmation.
Execution role	Amazon_EventBridge_SecurityHub_SNS_Alerts	Allows EventBridge to publish matched events to SNS.

Evidence

Figure 2 - SecurityHubAlerts Email Subscription Created

The screenshot shows the AWS SNS console interface for a subscription. At the top, a green success banner indicates the subscription was created successfully. Below the banner, the subscription name is partially visible as "Subscription: effb...". The "Details" section shows the following information:

- ARN:** [Redacted]
- Endpoint:** [Redacted]
- Topic:** SecurityHubAlerts
- Status:** Pending confirmation
- Protocol:** EMAIL

The "Subscription filter policy" section shows a policy named "Redrive policy (dead-letter queue)". A message states: "No filter policy configured for this subscription. To apply a filter policy, edit this subscription." with an "Edit" button.

The SNS evidence shows a success banner indicating the subscription to SecurityHubAlerts was created successfully. The topic is SecurityHubAlerts, protocol is EMAIL, and status is Pending confirmation at the time of capture. Sensitive ARN and endpoint values were redacted.

Figure 3 - EventBridge Rule Sends Security Hub Imported Findings to SNS

The screenshot shows the "Review and create" page for an EventBridge rule. The rule is named "SecurityHubFindings" and is in an "Enabled" status. The event bus is set to "default". The event pattern is defined as follows:

```
1 {
2   "source": ["aws.securityhub"],
3   "detail-type": ["Security Hub Findings - Imported"]
4 }
```

The target is configured to send events to the "SecurityHubAlerts" SNS topic. The role assigned is "Amazon_EventBridge_SecurityHub_SNS_Alerts". The input to the target is "Matched event".

Step 4: Configure tag(s) shows no tags associated with this resource.

The EventBridge review evidence shows rule name SecurityHubFindings, status Enabled, default event bus, standard rule type, event pattern matching source aws.securityhub and detail-type Security Hub Findings - Imported, and target SecurityHubAlerts as an SNS topic. The role shown is Amazon_EventBridge_SecurityHub_SNS_Alerts.

Security Analysis

This project converts passive Security Hub visibility into active notification by creating an EventBridge rule that reacts to new imported findings.

Routing through SNS is simple and appropriate for this lab because expected volume is low. It should not be treated as a mature enterprise alerting workflow.

The EventBridge rule sends all imported Security Hub findings. This is broad and useful for early validation, but production should usually filter by severity, source service, workflow status, or finding type.

The SNS subscription was pending confirmation at evidence capture time. Email delivery is not fully active until the recipient confirms the subscription.

Because the target is SNS, a future version can add or remove subscriptions without redesigning the EventBridge rule.

Lessons Learned

Events are different from logs. EventBridge handles near-real-time event matching, while logs are stored records for later analysis.

Security Hub imports findings from security services and emits imported-finding events that EventBridge can match.

EventBridge rules have two main parts: a pattern that decides what to match and a target that decides where to send the matched event.

Email alerting is useful for constrained labs, but high-volume security alerts need triage, deduplication, routing, and ownership workflows.

Production Improvements

- Confirm the SNS email subscription and capture evidence of confirmed status.
- Filter EventBridge notifications to high and critical findings first.
- Route findings to a SIEM, SOAR platform, ticketing system, or incident-response queue instead of raw email.
- Add a dead-letter queue for failed EventBridge-to-SNS delivery.
- Add input transformation to reduce email noise and highlight account, region, severity, title, and resource.
- Create separate rules for GuardDuty, Security Hub severity levels, and compliance findings.

Skills Demonstrated

Amazon EventBridge	Amazon SNS	AWS Security Hub
Event patterns	SNS email subscriptions	SecurityAudit operations
Finding alerting	Target routing	Near-real-time monitoring

Resume Bullet

RESUME

Built an event-driven AWS Security Hub alerting path by creating a SecurityHubAlerts SNS topic, subscribing an email endpoint, and configuring an EventBridge rule to forward imported Security Hub findings to SNS.

STAR Interview Story

Situation: Security Hub was enabled and centralized, but the environment did not yet notify the operator when new findings were imported.

Task: Create a simple lab alerting path from Security Hub findings to email without using the management account.

Action: I created an SNS topic and email subscription in SecurityAudit, then created an EventBridge rule matching Security Hub imported findings and targeting the SNS topic.

Result: The lab now has an event-driven notification path for Security Hub findings, ready for later filtering, severity routing, and SIEM or ticketing integration.

Version History

v1.0.0	July 16, 2026	Initial documented implementation. Created SecurityHubAlerts SNS topic and email subscription, created SecurityHubFindings EventBridge rule, matched Security Hub imported findings, targeted SNS email notification, documented pending subscription confirmation, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Use EventBridge for Security Hub Alerts.
- User-provided SNS SecurityHubAlerts email subscription screenshot.
- User-provided EventBridge SecurityHubFindings rule screenshot.
- Project 23 - Security Hub Central Configuration.